

UNIT. NO. 1

Introduction to Cloud Computing

- **Overview of Cluster Computing, Grid Computing, Parallel Computing & Distributed Computing**
- **Definition of cloud, Key Characteristics, Benefits and Risks in Cloud Computing, Economics of Cloud Computing**
- **Cloud Service Models, Cloud Deployment Models, Cloud Computing NIST Model, Cloud Computing Reference Architecture by IBM**
- **Infrastructure-as a-Service (IaaS), Platform-as-a Service (PaaS), Software-as-a-Service (SaaS), Overview of Amazon EC2, Google App Engine, Salesforce.com**

Overview of Parallel Computing Distributed Computing, Cluster Computing & Grid Computing

Parallel Computing

- Parallel computing is a type of computation where multiple calculations or processes are carried out simultaneously, enabling faster and more efficient processing of large datasets.
- It involves breaking down a problem into smaller, independent tasks that can be executed concurrently on multiple processors or computing units. This approach contrasts with serial computing, where tasks are executed one after another.
- Processing of multiple tasks simultaneously on multiple processors is called parallel processing. The parallel program consists of multiple active processes (tasks) simultaneously solving a given problem.
- A given task is divided into multiple sub tasks using a divide-and-conquer technique, and each sub task is processed on a different central processing unit (CPU). Programming on a multi-processor system using the divide-and- conquer technique is called parallel programming.

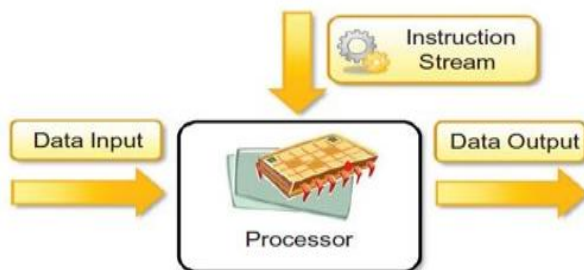
Key Concepts

Concept	Description
Task Decomposition	Breaking a problem into smaller tasks.
Concurrency	Tasks are logically simultaneous.
Synchronization	Coordination between parallel tasks.
Communication	Exchange of data among tasks or processors.

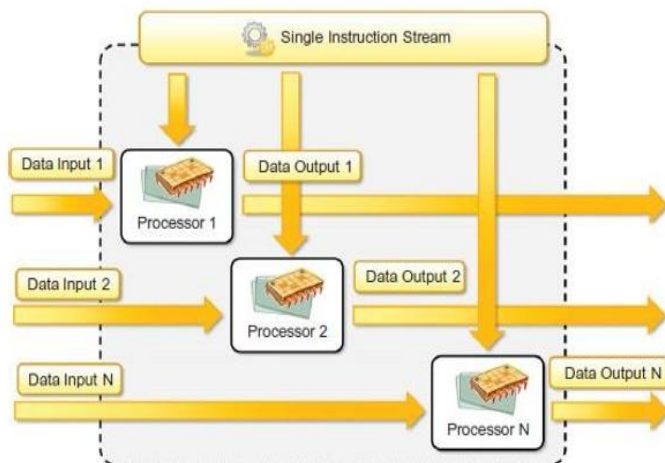
Flynn's Taxonomy

Model	Full Form	Description
SISD	Single Instruction Single Data	Traditional serial computing
SIMD	Single Instruction Multiple Data	Vector processors, GPUs
MISD	Multiple Instruction Single Data	Rare, fault tolerance systems
MIMD	Multiple Instruction Multiple Data	Most common (e.g., multicore CPUs, clusters)

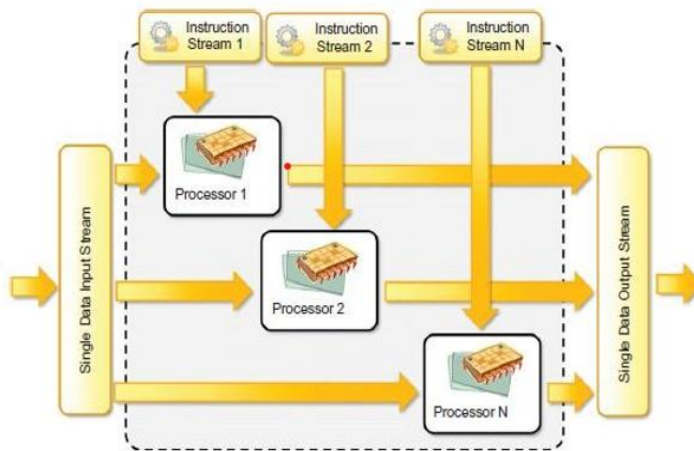
SISD



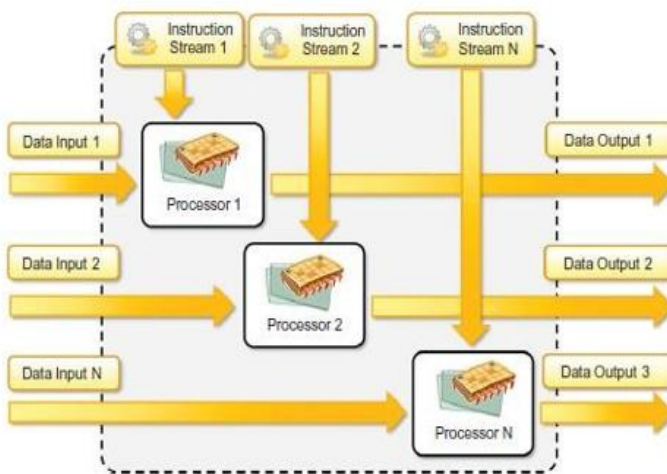
SIMD



MISD



MIMD



Distributed Computing

- A distributed system is a collection of independent computers that appears to its users as a single coherent system.
- This definition is general enough to include various types of distributed computing systems that are especially focused on unified usage and aggregation of distributed resources.
- Distributed computing is a method where multiple computers, or nodes, work together over a network to solve a problem by dividing the workload and processing it concurrently.
- This approach allows for increased performance, scalability, and resilience compared to relying on a single computer.

Key Concepts

- **Nodes:** Individual computers or servers that are part of the distributed system.
- **Network:** The communication medium that connects the nodes.
- **Message Passing:** The mechanism by which nodes communicate and coordinate their actions.
- **Parallelism:** The ability to execute multiple tasks simultaneously across different nodes.
- **Scalability:** The ability to easily add or remove nodes to adjust to changing workload demands.
- **Fault Tolerance:** The ability of the system to continue operating even if some nodes fail.

Working

- **Task Decomposition:** The overall problem is broken down into smaller, manageable tasks.
- **Task Distribution:** These tasks are then distributed across the available nodes in the network.
- **Parallel Processing:** Each node independently processes its assigned task.
- **Result Aggregation:** The results from individual nodes are then combined to produce the final solution.

Benefits of Distributed Computing

Increased Performance:

- By leveraging parallelism, distributed systems can significantly reduce processing time for large-scale computations.

Improved Scalability:

- Distributed systems can easily scale up or down to handle varying workloads, making them ideal for handling large data volumes and complex applications.

Enhanced Resilience:

- Distributed systems are more resilient to failures because the workload can be redistributed to other nodes if one node fails.

Cost-Effectiveness:

- Distributed systems can be built using commodity hardware, which can be more cost-effective than using a single, powerful machine.

Resource Sharing:

- Distributed systems allow for the sharing of resources like computing power, storage, and data across multiple machines.

Examples of Distributed Computing

Cloud Computing:

- Cloud platforms like AWS, Google Cloud, and Azure use distributed computing to provide on-demand access to computing resources.

Big Data Processing:

- Systems like Apache Hadoop and Spark use distributed computing to process massive datasets.

Web Services:

- Many web applications are built using distributed systems to handle large traffic and provide high availability.

Scientific Simulations:

- Distributed computing is used to simulate complex scientific phenomena, such as weather patterns or protein folding.

Blockchain Technology:

- Blockchain networks rely on distributed systems to maintain a shared, immutable ledger of transactions.

Cluster Computing

- Cluster computing is a collection of tightly or loosely connected computers that work together so that they act as a single entity.
- The connected computers execute operations all together thus creating the idea of a single system. The clusters are generally connected through fast local area networks (LANs)
- Cluster Computing is a form of parallel or distributed computing where a group of interconnected standalone computers (nodes) work together as a single system to provide high availability, reliability, and performance.

Type	Description	Example Use
High-Performance Clusters (HPC)	Focus on fast computation using parallel processing.	Weather simulations, scientific modeling
High-Availability Clusters (HA)	Ensure service continuity even if a node fails.	Banking systems, web hosting
Load Balancing Clusters	Distribute workloads evenly among nodes.	Web servers, application servers
Storage Clusters	Clustered for scalable and reliable data storage.	Data centers, cloud storage systems

Classification of Cluster:

1. Open Cluster:

- IPs are needed by every node and those are accessed only through the internet or web. This type of cluster causes enhanced security concerns.

2. Close Cluster:

- The nodes are hidden behind the gateway node, and they provide increased protection. They need fewer IP addresses and are good for computational tasks.

Advantages of Cluster Computing:

- High availability and fault tolerance
- Scalability by adding more nodes
- High performance via parallel processing
- Cost-effective use commodity hardware
- Flexibility and extensibility

Grid Computing

- Grid computing is a system that links geographically distributed computer resources to work together on a single, large task, acting as a virtual supercomputer.
- It involves connecting various computers, often across different locations, to tackle complex problems that would be difficult for a single machine to handle.
- Grid Computing is a type of distributed computing where resources from multiple locations—often across different administrative domains—are pooled together to solve large-scale tasks.
- It emphasizes resource sharing, heterogeneity, and geographical distribution, and is used for complex, compute-intensive problems.

Architecture of Grid Computing

Layered Architecture

- Fabric Layer – Access to local resources (CPU, memory, network, etc.)
- Connectivity Layer – Communication & authentication protocols (e.g., TCP/IP, SSL).
- Resource Layer – Resource allocation and control.
- Collective Layer – Resource discovery, scheduling, load balancing.
- Application Layer – User applications running on the grid.

Types

Type	Description	Example
Computational Grid	Focuses on CPU-intensive tasks.	Protein folding, simulations
Data Grid	Handles large-scale data storage and access.	CERN's LHC data analysis
Collaborative Grid	Facilitates data and app sharing among users.	Research projects
Utility Grid	On-demand resource provisioning.	Similar to cloud services

Benefits

- Increased Processing Power: Grid computing allows for handling complex tasks that would be impossible for a single computer.
- Cost-Effectiveness: It utilizes existing resources, potentially reducing the need for expensive, specialized hardware.
- Scalability: Grids can be scaled up or down by adding or removing computers as needed.
- Resource Optimization: It allows for efficient use of available computing resources.

Definition of cloud, Cloud Deployment Models, Cloud Service Models, Key Characteristics, Benefits and Risks in Cloud Computing

Definition of cloud computing:

- According to NIST (National Institute of Standards and Technology): “Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.”
- Cloud computing is the delivery of computing services – including servers, storage, databases, networking, software, analytics, and intelligence – over the internet (“the cloud”).
- It allows users to access and use these resources on demand, without the need to manage physical infrastructure themselves. Essentially, cloud computing provides computing power and storage as a service, enabling users to pay only for what they use, and scale resources as needed.

Key Features:

- On-demand self-service – Users can access computing resources without human interaction with the service provider.
- Broad network access – Services are available over the network and accessed through standard mechanisms.
- Resource pooling – Resources are pooled to serve multiple users using a multi-tenant model.
- Rapid elasticity – Capabilities can be quickly scaled up or down based on demand.
- Measured service – Usage is monitored, controlled, and reported for transparency.

Key Characteristics of Cloud Computing

Defined by NIST, the five essential characteristics of cloud computing are:

- On-Demand Self-Service
 - Users can automatically provision computing resources like storage or processing power without human interaction with the provider.
- Broad Network Access
 - Services are available over the network and accessible through standard devices like laptops, smartphones, and tablets.

- Resource Pooling
 - Providers pool resources to serve multiple customers using a multi-tenant model, dynamically assigning resources as needed.
- Rapid Elasticity
 - Resources can be quickly scaled up or down depending on demand, appearing unlimited to users.
- Measured Service
 - Cloud systems automatically control and optimize resource use by metering usage (e.g., storage, bandwidth, processing), providing transparency.

Benefits of Cloud Computing

Cost Efficiency

- Pay-as-you-go model reduces capital expenses (Capital Expenses) for hardware and software.

Scalability & Flexibility

- Easily scale resources up or down based on needs.

Accessibility & Mobility

- Access from anywhere with an internet connection.

Disaster Recovery & Backup

- Cloud providers offer data redundancy and recovery services.

Automatic Updates

- Service providers manage software and security updates automatically.

Environmental Efficiency

- Shared infrastructure reduces carbon footprint compared to dedicated data centers.

Risks in Cloud Computing

Data Security & Privacy

- Storing sensitive data on remote servers increases the risk of data breaches or unauthorized access.

Compliance Issues

- Organizations may face challenges meeting regulatory requirements like GDPR, HIPAA, etc.

Downtime & Service Outages

- Dependence on internet connectivity and provider uptime introduces risk.

Vendor Lock-In

- Difficulty in migrating between providers due to proprietary technologies and data formats.

Limited Control

- Customers have less control over infrastructure and may be restricted in custom configurations.

Insider Threats

- Employees of cloud providers might pose a threat if proper access controls aren't in place.

Economics of Cloud Computing

Cloud economics studies the financial impact, resource usage, and cost-benefit value of moving from physical data centers to cloud platforms. It replaces heavy upfront infrastructure investments with a flexible, pay-as-you-go operational model, allowing companies to optimize IT spending based on real-time business demand.

Financial Shifts and Models

- **CapEx to OpEx:** Swap large upfront hardware purchases for predictable, usage-based operational expenses.
- **Pay-as-you-go:** Pay only for the exact compute time, storage space, or network bandwidth you consume.
- **Pricing Options:** Use on-demand rates for flexibility or commit to 1-to-3-year reservations to secure discounts of up to 72%.

Value and Efficiency Drivers

- **Elasticity:** Scale resources up during busy times and down when quiet to avoid paying for idle hardware.
- **Labor Savings:** Remove the need for local IT teams to manually install, repair, and cool physical servers.
- **Speed to Market:** Launch new apps and features faster, boosting overall business productivity and revenue growth.

Risks and Cost Management

- **Hidden Expenses:** Unmanaged cloud environments often run up surprise bills through forgotten data or runaway workloads.
- **Migration Costs:** Moving legacy applications overtakes time, planning, and specialized engineering work.
- **FinOps Discipline:** Companies need active cost tracking and budgeting tools to keep cloud waste under control

Economical background of the cloud is more useful for developers in the following ways:

- Pay as you go model offered by cloud providers.
- Scalable and Simple.

There are three different **Pricing Strategies** that are introduced by Cloud Computing: Tiered Pricing, Per-unit Pricing, and Subscription-based Pricing. These are explained as following below.

1. **Tiered Pricing:** Cloud Services are offered in the various tiers. Each tier offers to fix service agreements at a specific cost. Amazon EC2 uses this kind of pricing.
2. **Per-unit Pricing:** The model is based upon the unit-specific service concept. Data transfer and memory allocation include in this model for specific units. GoGrid uses this kind of pricing in terms of RAM/hour.
3. **Subscription-based Pricing:** In this model, users are paying periodic subscription fees for the usage of the software.

A common cloud economics approach involves breaking the analysis across the following four pillars:

- **Total cost of ownership:** A data center TCO analysis includes infrastructure costs, such as the purchase or lease of the physical building, power and cooling systems, and networking. Additionally, there are costs associated with hardware and software; personnel, including salaries and benefits for IT staff; maintenance and support, including software updates; and energy given that data centers require a substantial amount of power to operate.
- In a cloud model, the TCO shifts to monthly usage costs for compute and storage and outlines how a shift in staffing needs will affect payroll. A cloud economics analysis breaks these details down, providing cost/benefit insights across the organization.
- **Efficiency improvements:** Cloud platforms provide organizational benefits, including improved collaboration and stronger DevOps support. Efficiency improvements can vary in size and scope; some examples include faster data sharing between sales and marketing teams, new IT innovations after freeing staff from data center responsibilities, and automatic data integration into applications such as self-service analytics.

- **Operational resilience:** Local data centers face a constant risk of real-world issues affecting availability. Natural disasters, power outages, and even local politics regarding infrastructure can all cause unexpected downtime for on-premises equipment. Cloud providers build uptime guarantees into service level agreements and design their business models around redundancy, giving customers greater resilience.
- **Operational agility:** Many different factors can change the volume of network traffic, including app updates and remote access. To absorb these spikes, on-premises data centers need the physical resources to handle peak loads. In a cloud environment, scalability becomes as simple as upgrading to a higher service tier. This creates organizational agility, whether spikes are temporary, such as with holiday sales, or the start of significant and lasting growth.

The total economic benefit of cloud migration includes immediate cost savings and the financial impact of its many functional benefits across an organization. The following are a few ways organizations reduce costs with the cloud:

- **Reduced TCO:** With an on-premises configuration, an organization must budget for all stages of a hardware cycle, from initial acquisition to maintenance costs and configuration time. In a cloud environment, the provider handles data center needs. The provider's staff takes care of replacements and repairs, relieving customers of the need for capital investments and ongoing licensing and maintenance costs.
- **Save on intangibles:** If your IT staff aren't burdened by maintenance tasks, such as troubleshooting a problematic server, where could they spend that time and effort? Moving to the cloud frees them up so they can further IT innovation for the organization and drive improvements that create efficiencies and stability.
- **OpEx instead of CapEx:** By moving to the cloud, organizational budgets shift from a mainly CapEx model to an OpEx model. This creates a more efficient spending strategy as cloud-based OpEx budgeting doesn't require the long-term up-front investments in compute, storage, and capacity—all resources that may never be fully utilized.
- **Scale as needed:** When working with an on-premises data center, hardware plans must account for the maximum possible resource usage. For example, if an upcoming launch is expected to drive traffic, then IT must scale to that volume with additional safety-net coverage. However, if the app's launch drives less than the anticipated traffic, all that overhead will be wasted. Cloud providers scale on demand with a pay-as-you-go pricing model.

Cloud Deployment Models:

- A cloud deployment model fundamentally determines where the infrastructure for your deployment exists and who owns and controls that infrastructure. It also figures out the nature and intention of the cloud.
- Cloud deployment models define how cloud services are made available to users.
- The four main models are:
- Public Cloud, Private Cloud, Hybrid Cloud, and Community Cloud.

Public Cloud

- Cloud infrastructure is owned and operated by third-party providers (e.g., AWS, Microsoft Azure, Google Cloud).
- Available to the public over the internet.
- Advantages:
 - Cost-effective (pay-as-you-go)
 - Easy scalability
 - No maintenance required by the user
- Disadvantages:
 - Less control over data
 - Potential security and privacy concerns

Private Cloud

- Cloud infrastructure is used exclusively by a single organization. It can be managed internally or by a third party.
- Restricted to the organization.

Advantages:

- High security and control
- Customizable architecture

Disadvantages:

- Higher costs
- Requires in-house expertise for maintenance

Hybrid Cloud

- A combination of public and private clouds, allowing data and applications to move between them.

- Both public and private access depending on the service or data.

Advantages:

- Flexibility and scalability
- Cost efficiency with better control over sensitive data

Disadvantages:

- Complex management
- Potential security challenges in integration

Community Cloud

- Cloud infrastructure is shared by multiple organizations with common goals or compliance requirements (e.g., government, healthcare).
- Only the participating organizations.

Advantages:

- Shared cost among similar users
- Collaborative environment

Disadvantages:

- Limited scalability compared to public cloud
- Governance and management challenges

Cloud service models

- Cloud service models define what kind of services are provided to the users over the cloud. These models determine the level of control, flexibility, and management required by the user.
- Cloud Computing means using the internet to store, manage, and process data instead of using your own computer or local server.
- The data is stored on remote servers, that are owned by companies called cloud providers such as Amazon, Google, Microsoft). These companies charge you based on how much you have used their services.

Infrastructure as a Service (IaaS)

- Provides virtualized computing resources over the internet—like servers, storage, and networking.
- User Controls: Operating system, storage, deployed applications.
- Provider Manages: Physical infrastructure and virtualization.
- Examples: Amazon EC2, Microsoft Azure VM, Google Compute Engine

- Use Case: Hosting websites, storage backup, test and development environments.

Platform as a Service (PaaS)

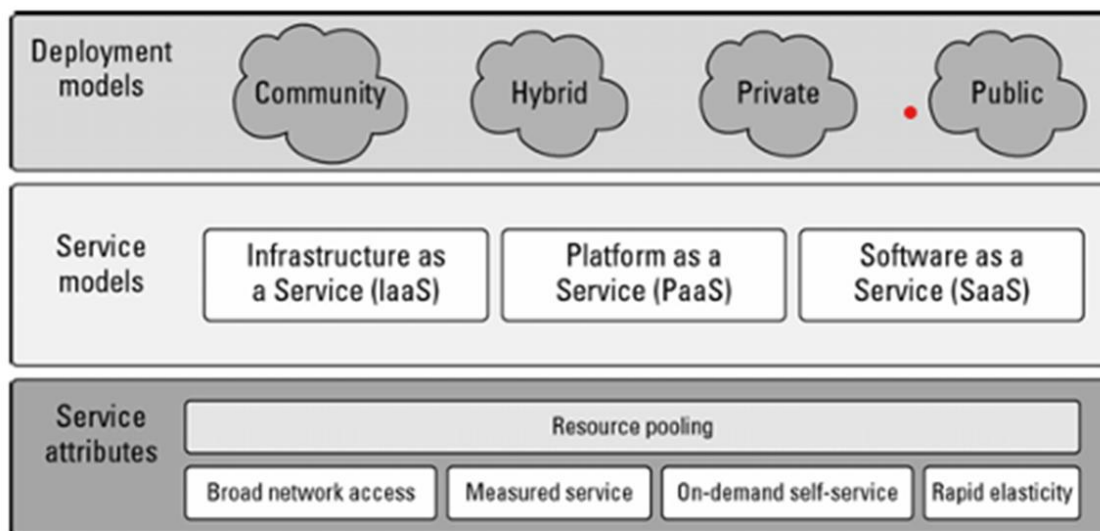
- Provides a platform or environment to develop, test, and deploy applications without managing the underlying hardware or OS.
- User Controls: Applications and data.
- Provider Manages: Servers, OS, networking, storage.
- Examples: Google App Engine, Microsoft Azure App Services, Heroku
- Use Case: Application development and deployment.

Software as a Service (SaaS)

- Delivers fully functional software applications over the internet on a subscription basis.
- User Controls: Just the application configuration (limited).
- Provider Manages: Everything—applications, data, OS, infrastructure.
- Examples: Google Workspace (Gmail, Docs) Microsoft 365 Salesforce
- Use Case: Email, CRM, collaboration tools.

Cloud Computing NIST Model

- The United States government is a major consumer of computer services and, therefore, one of the major users of cloud computing networks.
- The U.S. National Institute of Standards and Technology (NIST) has a set of working definitions that separate cloud computing into service models and deployment models.



The NIST Definition of Cloud Computing

- Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.
- This cloud model is composed of five essential characteristics, three service models, and four deployment models.

Deployment Models

Private cloud:

- The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.

Community cloud:

- The cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.

Public cloud:

- The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.

Hybrid cloud:

- The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

Service Models

Software as a Service (SaaS).

- The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure.
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user specific application configuration settings.

Platform as a Service (PaaS).

- The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider.
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.

Infrastructure as a Service (IaaS).

- The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications.
- The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., host firewalls).

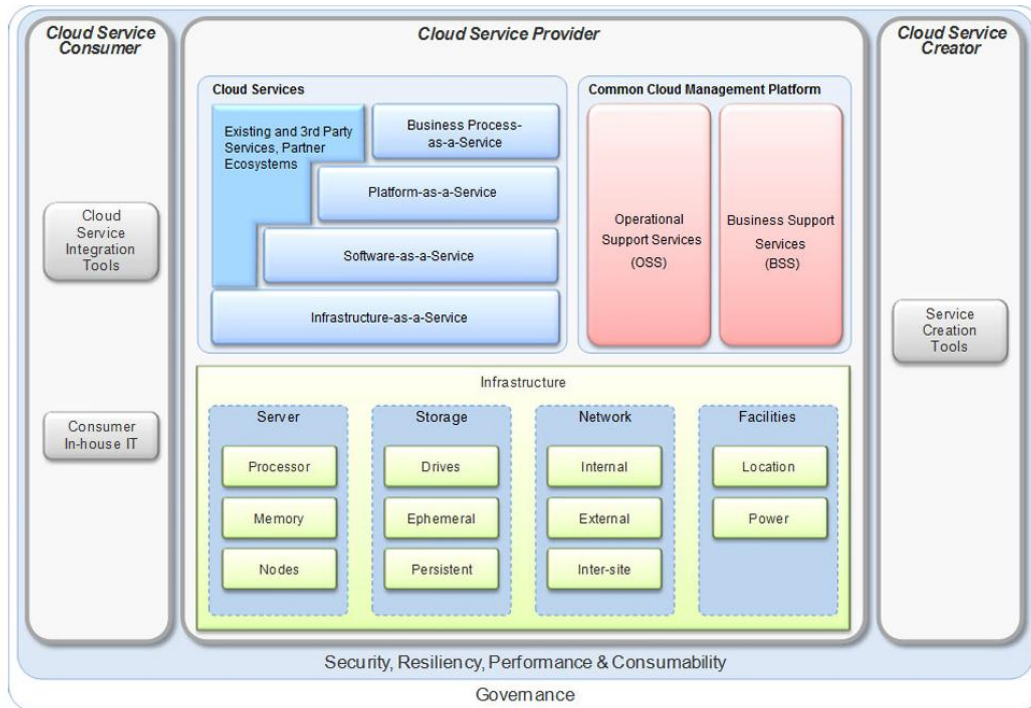
Essential Characteristics:

- **On-demand self-service:** A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
- **Broad network access:** Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).
- **Resource pooling:** The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
- **Rapid elasticity:** Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.
- **Measured service:** Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts).

The Cloud Computing Reference Architecture (CCRA) by IBM

- IBM defined a reference architecture for the cloud computing domain and named it the IBM Cloud Computing Reference Architecture. It is an evolving architecture,

based on real-world input from many cloud implementations around the globe, and was submitted to the Open Group Cloud Architecture Project.



- **As shown in Figure, the IBM CCRA defines the following interrelated roles:**
- Cloud Service Creator
- Cloud Service Provider
- Cloud Service Consumer
- These roles are interrelated in that a Cloud Service Creator is responsible for creating a cloud service, which can be run by a Cloud Service Provider, and exposed to Cloud Service Consumers.
- Multiple roles may be fulfilled by the same organization or person.

Cloud Service Creator

- The Cloud Service Creator is responsible for creating a cloud service. The creator may be an individual or an organization that designs, implements, and maintains runtime and management artifacts that are specific to a cloud service.
- Typically, Cloud Service Creators build their cloud services by using functionality that is exposed by a Cloud Service Provider. Also typical is that the operations staff, who are responsible for operating a cloud service, are closely integrated with the development organization that develops the service (this integration is commonly referred to as DevOps).

- This close integration helps to achieve the delivery efficiency that is expected from cloud services, because it allows a short feedback loop to implement changes in the cloud service.

Cloud Service Provider

- The Cloud Service Provider has the responsibility of providing cloud services to Cloud Service Consumers.

The provider sets up the cloud service, and manages the effective running of the service, which can include the following tasks (and others):

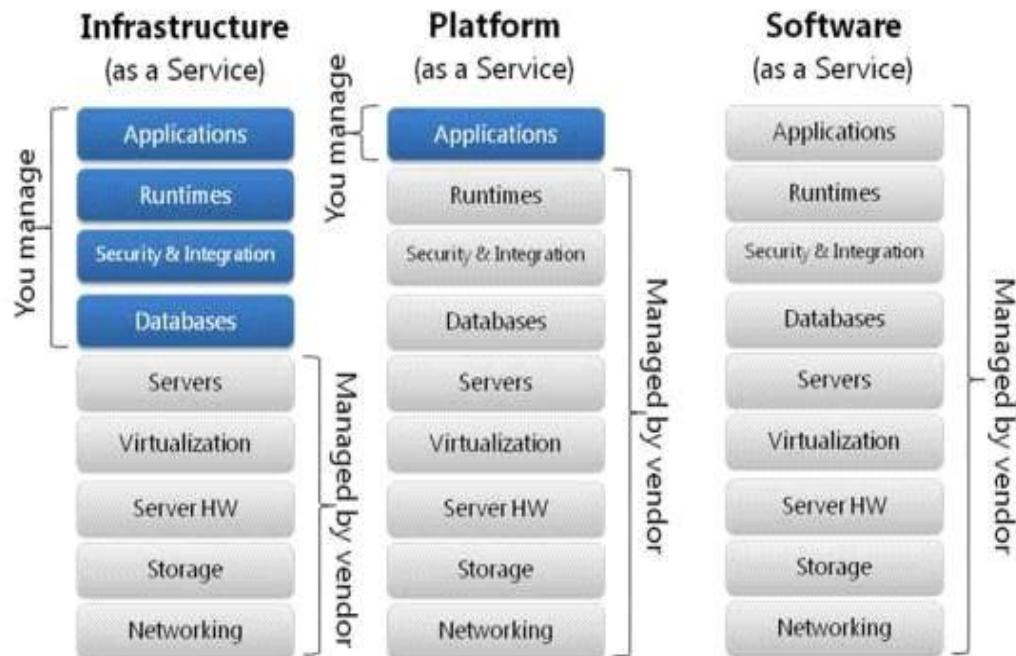
- Determine performance service levels and management strategies.
- Monitor performance of virtualization infrastructure and service level agreements (SLAs).
- Manage long-term capacity and performance trends.
- Analyze how to prevent costly service quality problems.
- Ensure alignment of business and operational support systems.
- Track performance against the provider's business plan.

A Cloud Service Provider might be a link within a chain of service providers and service consumers, with each provider adding some value to the service within the chain. In this case, each service provider needs to establish a partnership with their service provider, to be able to guarantee service levels to their clients.

Cloud Service Consumer

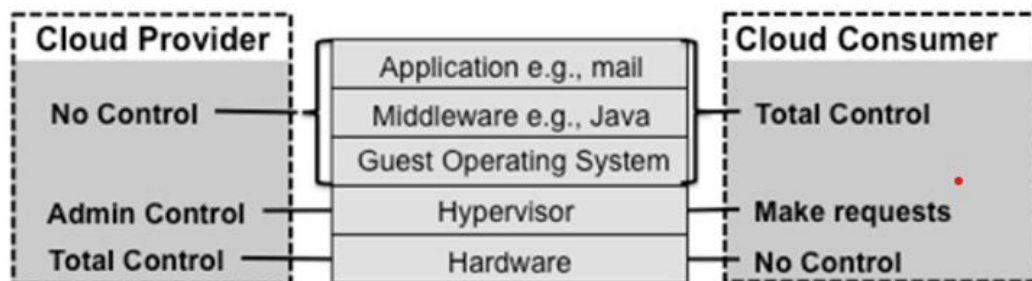
- A Cloud Service Consumer is the user of a cloud service.
- The consumer might be an organization, a human being, or an IT system that requests, uses, and manages instances of a cloud service.
- Managing a service can include performing activities such as changing quotas for users, changing CPU capacity assigned to a virtual machine (VM), or increasing maximum number of seats for a web conference.
- The service consumer may be billed for all (or a subset) of its interactions with the cloud service and the provisioned service instances.
- Within the Cloud Service Consumer role, more specific roles might exist. The consumer organization might require a technical role responsible for making service consumption work from a technical perspective.
- There might also be a businessperson on the consumer side who is responsible for the financial aspects of consuming the service. In simple public cloud scenarios, all these consumer roles could be collapsed into a single person.
- The Cloud Service Consumer browses the service offering catalog and triggers service instantiation and management from there.

Infrastructure-as-a-Service (IaaS), Platform-as-a Service (PaaS), Software-as-a-Service (SaaS)

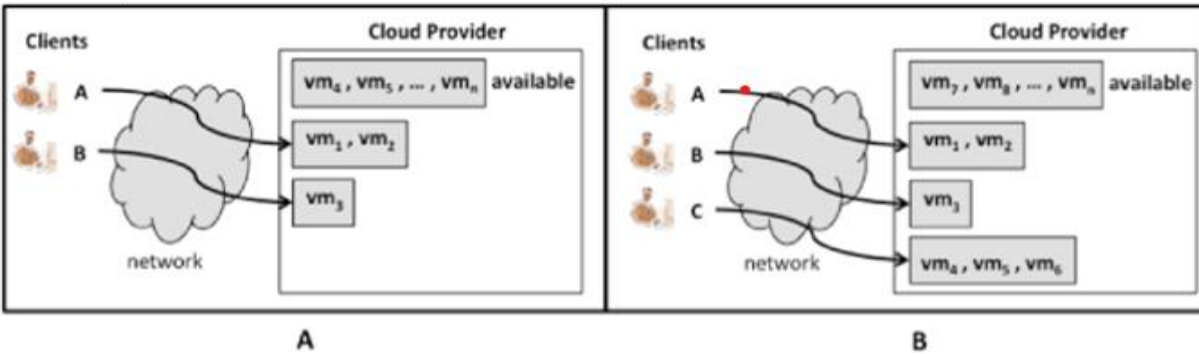


INFRASTRUCTURE-AS-A-SERVICE (IAAS)

- IaaS provides virtual machines, virtual storage, virtual infrastructure, and other hardware assets as resources that clients can provision.
- Most large Infrastructure as a Service (IaaS) providers rely on virtual machine technology to deliver servers that can run applications.



IaaS Component Stack and Scope of Control

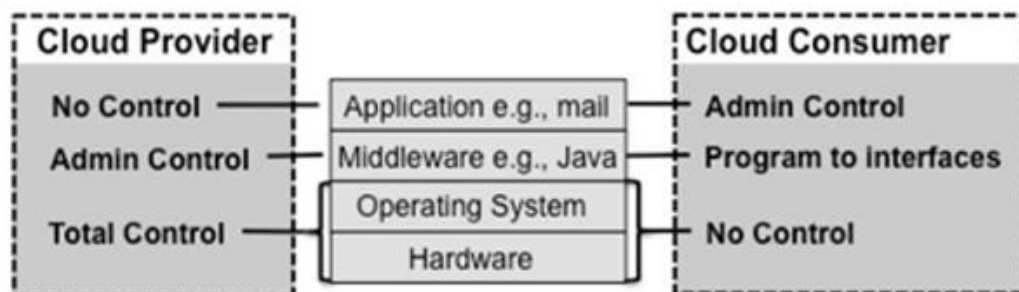


IaaS Provider/Consumer Interaction Dynamics

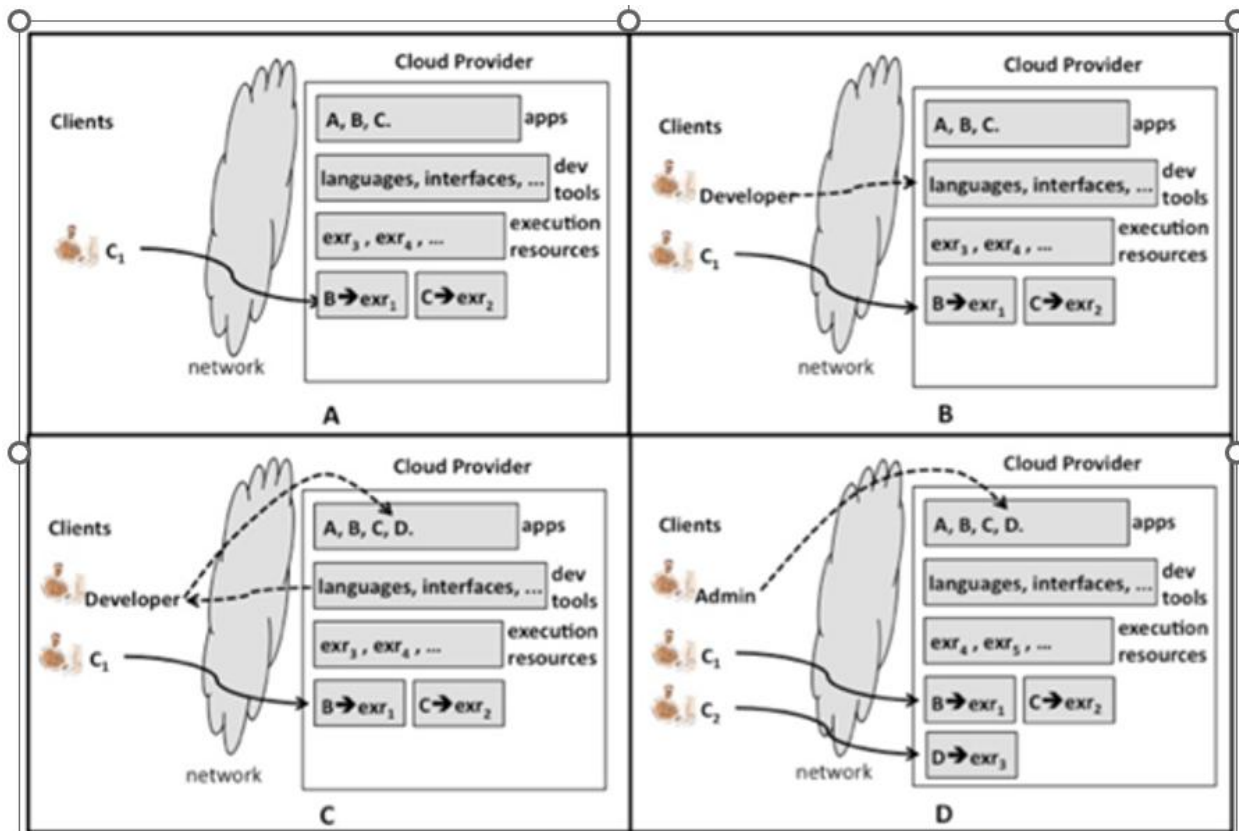
- The figure A Shows client that are interacting with an IaaS cloud over a network.
- The provider has several available virtual machines that he can allocate to clients.
- In the figure, client A has access to vm_1 and vm_2 , and client B has access to vm_3 .
- The provider retains vm_4 through vm_n , where it is presumed that n is larger than the number of VMs any client is expected to request.
- Another situation is shown on figure B, where just after a new client C has requested and acquired access to three more vms.
- At this stage, client C has access to vm_4 , vm_5 , and vm_6 , and the provider now retains only vm_7 through vm_n .

PLATFORM-AS-A-SERVICE (PAAS)

- The PaaS model makes all the facilities required to support the complete life cycle of building and delivering web applications and services entirely available from the Internet, all with no software downloads or installation for developers, IT managers, or end users.



PaaS Component Stack and Scope of Control

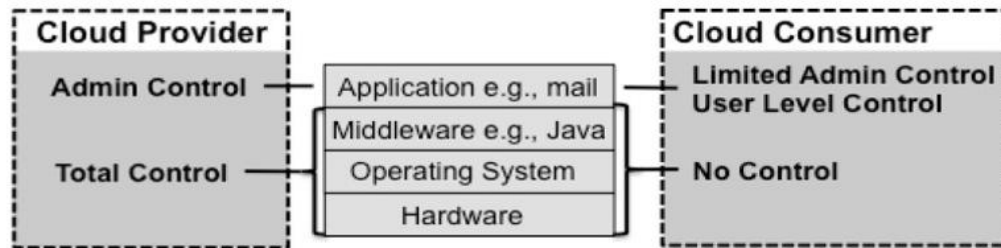


PaaS Provider/Consumer Interaction Dynamics

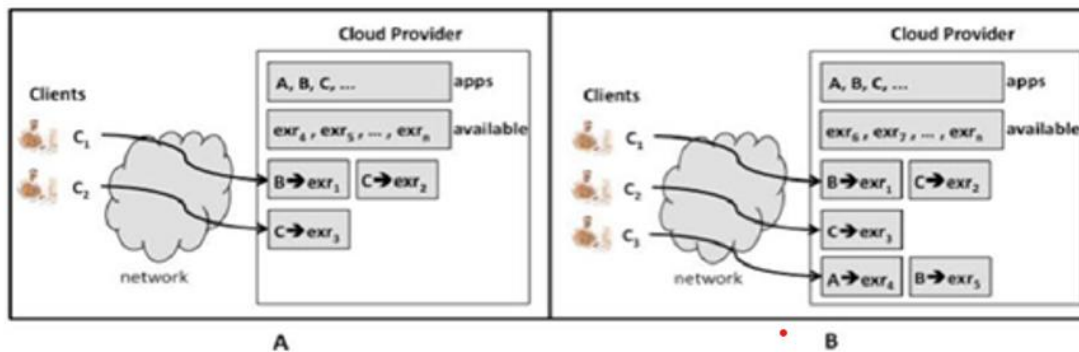
- Figure A shows the PaaS provider that has a current inventory of three applications deployed (apps), set of development tools (dev tools), and a set of execution environments (exr). There are also depicted two active applications, B → exr₁ and C → exr₂ indicating that applications B and C are using separate execution resources.
- Figure B shows the developer client accessing the development tools of the provider. An example of such tools could be programming languages, compilers, interfaces, testing tools, and tools for deployment of an application.
- In figure C it can be seen how the developer uses the tools.
- Thus, one may download tools and use them locally in the developer 's infrastructure or can access those tools in the provider 's infrastructure as well.
- In each case the result is a new application D, as depicted in the figure, which is deployed onto the provider 's infrastructure.
- In figure D an administrator can be observed configuring the new application that has been made available, as well as a new client, C₂, using that new application.

SOFTWARE-AS-A-SERVICE (SAAS)

- SaaS, which is a Software deployed as a hosted service and accessed over the Internet.



SaaS Component Stack and Scope of Control



SaaS Provider/Consumer Interaction Dynamics

- Figure A represents a cloud providing services to two clients, C1 and C2.
- In a private cloud, the clients will belong to (or be associated with) a single consumer organization; in other deployment models the clients may represent different consumers.
- Abstractly, the cloud provider possesses a set of software applications ("apps" in the figure) that it is offering to the clients for use over the network.
- Moreover, the cloud provider manages application execution resources ("exr" in the figure).
- In Figure A, client C1 is currently using two applications, B and C.
- To execute the apps for client C1, the cloud provider has allocated two execution resources, exr1 and exr2, with exr1 supplying the processing power and other resources to run the B application.

- An execution resource could be, e.g., a physical computer, a virtual machine, or a running server program that can serve client requests, start a virtual machine, or even rent computing cycles and storage from another organization.
- Similarly, client C2, is using one application, C, which is supported by execution resource exr3. It should be noticed that the same application (C in this case) can be rented out to multiple clients at the same time, if the cloud provider can provide the execution resources to support the application.
- As shown in Figure B, when an additional client requests applications from the cloud, the cloud provider allocates extra execution resources to support the requested applications.

Overview of Amazon EC2, Google App Engine, Salesforce.com

Overview of Amazon EC2

- Amazon EC2 (Elastic Compute Cloud) is a web service provided by Amazon Web Services (AWS) that offers resizable compute capacity in the cloud.
- It allows users to run virtual machines (instances) on demand, making it easier to deploy and scale applications.
- EC2 encourages scalable deployment of applications by providing a web service through which a user can boot an Amazon Machine Image (AMI) to configure a virtual machine, which Amazon calls an "instance", containing any software desired.
- A user can create, launch, and terminate server-instances as needed, paying by the second for active servers – hence the term "elastic". EC2 provides users with control over the geographical location of instances that allows for latency optimization and high levels of redundancy.

Key Features of Amazon EC2

- Scalability: Launch and terminate instances as needed.
- Variety of Instance Types: Optimized for compute, memory, storage, or GPU.
- Elastic Load Balancing (ELB): Distribute traffic across instances.
- Auto Scaling: Automatically increase or decrease capacity.
- Custom AMIs (Amazon Machine Images): Create reusable templates for instances.
- Security: Integrated with AWS IAM, security groups, and key pairs.

Flexible Pricing Models:

- On-Demand Instances – pay for compute capacity by the hour or second.
- Reserved Instances – reserve instances at a lower price for 1–3 years.
- Spot Instances – bid for unused capacity at steep discounts.

Basic Components

- Instances: Virtual servers running applications.
- AMI (Amazon Machine Image): Pre-configured templates for launching instances.
- EBS (Elastic Block Store): Persistent block storage for EC2 instances.
- Security Groups: Virtual firewalls for controlling inbound/outbound traffic.
- Key Pairs: SSH authentication for Linux or Windows login.

Use Cases:

- Hosting websites and applications
- Running big data analytics and ML workloads
- High-performance computing (HPC)
- Dev/Test environments
- Disaster recovery and backups

EC2 Working:

- Choose an AMI.
- Select an instance type.
- Configure instance details (e.g., networking, storage).
- Add storage and tags.
- Configure security group (firewall rules).

Launch the instance and connect via SSH or RDP.

Overview of Google App Engine

- Google App Engine (GAE) is a fully managed, serverless Platform as a Service (PaaS) offered by Google Cloud.
- It enables developers to build, deploy, and host scalable web applications and mobile backends on Google's infrastructure without needing to manage the underlying servers or infrastructure.
- Google App Engine (also referred to as GAE or App Engine) is a cloud computing platform used as a service for developing and hosting web applications.
- Applications are sandboxed and run across multiple Google-managed servers.
- GAE supports automatic scaling for web applications, allocating more resources to the web application as the amount of requests increases.
- It was released as a preview in April 2008 and launched officially in September 2011.

Key features of Google App Engine include:

- **Serverless Architecture:**
- Developers can focus on writing code while App Engine handles server provisioning, patching, and scaling based on demand.
- **Automatic Scaling:**
- Applications automatically scale up or down in response to traffic fluctuations, ensuring high performance and availability.
- **Language Support:**
- App Engine supports various popular programming languages, including Python, Java, Node.js, Go, Ruby, PHP, and .NET, with custom runtimes available for other languages.
- **Managed Environment:**
- Google manages the infrastructure, allowing developers to concentrate on application development.
- **Built-in Services:**
- It provides integrated services like authentication, caching, monitoring, and data storage.
- **Versioning and Traffic Splitting:**
- Developers can deploy new versions of applications without disrupting existing ones and gradually roll out changes using traffic splitting.
- **Security Features:**
- Includes firewalls, encryption, and integration with Identity and Access Management (IAM) for granular control over access and traffic.

Overview of Salesforce

- Salesforce is a cloud-based software company specializing in customer relationship management (CRM).
- It provides a suite of applications and services designed to help businesses manage and improve their interactions with customers and prospects. These tools span various areas like sales, customer service, marketing, e-commerce, and analytics.

Core Services & Products

Sales Cloud

- Manages sales processes, leads, opportunities, and accounts.

Service Cloud

- Customer support and case management tools.

Marketing Cloud

- Automates marketing campaigns via email, social, mobile, and ads.

Commerce Cloud

- E-commerce and shopping experience management.

Experience Cloud (formerly Community Cloud)

- Build portals, forums, and customer communities.

Analytics Cloud (Tableau CRM)

Advanced data visualization and reporting.

Key Features

- Cloud-based and accessible from anywhere
- Highly customizable with point-and-click tools and coding
- Workflow automation and process builders
- Third-party integrations (Google, Microsoft, AWS, etc.)
- Mobile access through Salesforce mobile app
- Role-based access control and strong security model